

Boundary Information Geometry for S-box

Side-Channel Leakage: Walsh Covariance, Active

Fisher Structure, and CPA Belief Dynamics

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Abstract

Side-channel attacks on lightweight block ciphers often reduce locally to distinguishing key hypotheses through leakage generated by bijective S-boxes. This paper develops an information-geometric model for these finite key-induced distributions while separating three security-relevant spaces: the singular distributions induced by exact key hypotheses, the regular full-support exponential family used for Fisher geometry, and the adversary's posterior belief simplex. For an s -bit bijective S-box S and key hypothesis k , the exact distribution $P_k(a, b) = 2^{-s} \mathbf{1}[b = S(a \oplus k)]$ is supported on one graph inside $\{0, 1\}^{2s}$ and therefore lies on the boundary of the probability simplex. Walsh characters give the key-translation rule $\hat{P}_k(\alpha, \beta) = (-1)^{\alpha \cdot k} \hat{P}_0(\alpha, \beta)$. Using this rule as a structural input, we derive closed-form boundary covariance matrices and prove that, for every bijective S-box, their active subspace has dimension $2^s - 1$ and active eigenvalues equal 2^s in the unnormalised Walsh basis. We then interpret correlation power analysis under Gaussian leakage as Bayesian dynamics on the key-belief simplex. The Hamming-weight variance $s/(4\sigma^2)$ is a bijectivity-driven baseline, whereas pairwise likelihood gaps determine key-hypothesis discrimination and worst-case posterior-odds growth. Reproducible computations for PRESENT,

GIFT, and SKINNY show that S-boxes with the same Walsh-amplitude distribution can still have different hard and easy key differences under Hamming-weight CPA, while sharing the boundary covariance invariants forced by bijectivity.

Keywords: cybersecurity, side-channel analysis, correlation power analysis, S-box, information geometry, Fisher–Rao metric, Walsh–Hadamard transform, Bayesian inference

1 Introduction

1.1 Motivation

Side-channel attacks exploit implementation leakage rather than weaknesses in the abstract cipher alone. In many attacks on lightweight block ciphers, the local statistical object is a keyed S-box computation: an adversary observes or controls an input nibble or byte, measures a noisy leakage trace, and compares key hypotheses through predicted intermediate values. Correlation power analysis (CPA) is the standard example: each trace updates the adversary’s belief about the key using a leakage model such as Hamming weight.

The exact distribution induced by a keyed bijective S-box is not a smooth interior point of a probability simplex. For an s -bit bijective S-box $S : \{0, 1\}^s \rightarrow \{0, 1\}^s$ and key hypothesis $k \in \{0, 1\}^s$, the pair (a, b) with uniform a and $b = S(a \oplus k)$ is supported on only 2^s points of the 2^{2s} -point input-output space. Hence P_k is a singular boundary distribution. This matters because ordinary Fisher–Rao tensors, geodesics, and dual-flat coordinates are defined on full-support statistical models, not directly at support-deficient endpoints.

This paper develops a boundary information-geometric framework for this side-channel setting. The central discipline is to keep separate three objects that are often conflated in informal geometric descriptions:

(i) the regular full-support Walsh exponential family, where ordinary Fisher geometry	093
is valid;	094
(ii) the finite boundary orbit of exact key-induced S-box distributions;	095
(iii) the adversary's posterior belief simplex over key hypotheses, where CPA updates	096
take place.	097
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This separation makes it possible to use information geometry in a controlled way:	103
boundary distributions are represented through covariance limits, whereas Bayesian	104
key recovery remains a regular trajectory on the belief simplex as long as the prior	105
has full support and the leakage likelihood is positive.	106
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For cybersecurity and hardware-security evaluation, the practical output of this	109
separation is not a new S-box ranking by a single scalar. It is a diagnostic workflow	110
for identifying which quantities are unavoidable consequences of bijectivity, which	111
quantities are artefacts of the chosen Walsh coordinates, and which quantities actually	112
affect CPA discrimination. In particular, for Hamming-weight leakage the scalar	113
variance is identical for all bijective S-boxes of the same width, but the pairwise gaps	114
between key hypotheses need not be identical. Those gaps determine the expected	115
posterior-odds growth against wrong keys and therefore identify the key differences	116
that are locally hardest or easiest for a CPA adversary to reject.	117
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1.2 Relation to prior work	130
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The information-geometric background is the Fisher–Rao and dual-affine geometry	133
of finite statistical models and regular exponential families [1–3]. The closest coding-	134
theoretic precedent is the work of Ikeda, Tanaka, and Amari on LDPC and turbo	135
codes [4]: parity-check statistics define a regular information-geometric model and	136
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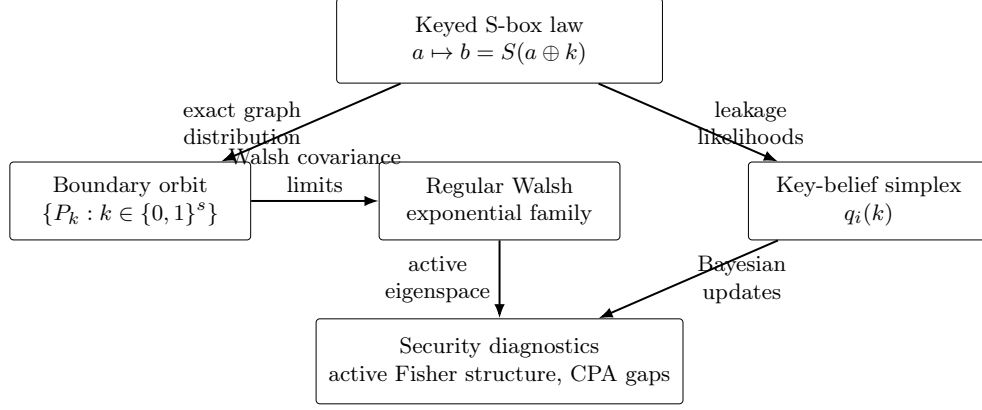


Fig. 1 Boundary-to-belief framework. The exact keyed S-box laws lie on a singular boundary orbit, the Fisher calculations are made in a regular Walsh exponential family or in boundary covariance limits, and CPA updates evolve on the separate key-belief simplex.

decoding trajectories are interpreted through projections. Here, the sufficient statistics are Walsh–Hadamard characters on S-box input-output pairs, and the inference trajectory is the CPA posterior over key hypotheses.

On the cryptographic side, the work is related to classical Walsh/LAT and DDT analysis of vectorial Boolean functions and S-boxes, including Nyberg’s differential uniformity [5]. Four-bit S-boxes and optimal 4-bit permutations have been classified in the Boolean-function literature, notably by Leander and Poschmann [6]. Recent geometric viewpoints in symmetric cryptanalysis also include geometric formulations of linear and differential cryptanalysis [7, 8]. The present paper does not propose a new classification of 4-bit S-boxes and does not claim that standard S-boxes with the same Walsh spectrum can be distinguished by spectral invariants. Instead, it asks a different question: what is the correct information-geometric object associated with exact keyed S-box distributions, and how should that object be connected to side-channel belief updates?

On the side-channel side, the leakage model and posterior update are consistent with differential and correlation power analysis [9, 10] and with information-theoretic analyses of key recovery attacks [11, 12]. The present framework is not a new attack

Table 1 Relation to existing metrics.

Metric or framework	Standard role	Role in this paper
Walsh/LAT spectrum	Measures linear correlations of vectorial Boolean functions and supports nonlinearity analysis.	Provides coordinates for the key-induced boundary orbit and for closed-form covariance diagnostics.
DDT and differential uniformity	Measures differential propagation and classical S-box resistance to differential attacks [5].	Used as part of the surrounding S-box context, but not replaced or re-ranked by the proposed geometry.
4-bit S-box classification	Classifies small permutations up to established equivalences [6].	Treated as prior knowledge; the paper does not claim a new classification or a new optimality criterion.
CPA and DPA likelihood models	Explain key recovery from noisy implementation leakage [9, 10].	Interpreted as Bayesian dynamics on the key-belief simplex with pairwise likelihood gaps.
Mutual information and key-rank measures	Quantify leakage and attack success at the experiment or trace-distribution level [11, 12].	Complemented by local structural diagnostics for keyed S-box laws and posterior-odds growth.
Information geometry of codes	Studies decoding and projections in regular statistical models [4].	Adapted to singular S-box graph distributions by separating boundary covariance, regular interiors, and belief-simplex updates.

and not a countermeasure. It is a formal model for separating baseline invariants forced by bijectivity from pairwise leakage gaps that control key-hypothesis discrimination.

Table 1 summarizes the intended position of the framework relative to standard S-box and side-channel criteria.

1.3 Setting and notation

Throughout, $s \geq 1$ denotes the S-box width in bits, $S : \{0, 1\}^s \rightarrow \{0, 1\}^s$ is a fixed bijection, $k \in \{0, 1\}^s$ is a key hypothesis, and $a \in \{0, 1\}^s$ is a uniformly distributed plaintext nibble or byte. All additions inside bit strings are over $\mathbb{F}_2$. The inner product is $\alpha \cdot a = \bigoplus_{i=1}^s \alpha_i a_i$. For $(\alpha, \beta) \in \{0, 1\}^s \times \{0, 1\}^s$, the Walsh character is

$$\chi_{\alpha\beta}(a, b) = (-1)^{\alpha \cdot a \oplus \beta \cdot b}.$$

We write $N = 2^s$ and $\mathcal{X} = \{0, 1\}^s \times \{0, 1\}^s$.

231 1.4 Contributions

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233 **C1. Boundary model for key-induced S-box distributions.** We model the exact
 234
 235 keyed S-box law

$$236 P_k(a, b) = 2^{-s} \mathbf{1}[b = S(a \oplus k)]$$

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238 as a boundary distribution on the input-output simplex. The Walsh translation
 239
 240 rule

$$241 \hat{P}_k(\alpha, \beta) = (-1)^{\alpha \cdot k} \hat{P}_0(\alpha, \beta)$$

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243 is used as a coordinate identity for the finite key orbit, not as a standalone
 244
 245 cryptanalytic novelty.

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 247 **C2. Closed-form boundary covariance and active Fisher structure.** We derive
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 249 the boundary covariance form

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$$252 \mathcal{I}_{ij}^\partial(k) = \text{Cov}_{P_k}(T_i, T_j)$$

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255 in Walsh coordinates and prove that, for every bijection $S : \{0, 1\}^s \rightarrow \{0, 1\}^s$, this
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 257 matrix has rank $2^s - 1$ and active eigenvalues all equal to 2^s in the unnormalised
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 259 Walsh basis. Thus the active boundary geometry is a structural property of
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 261 bijective S-box graphs, rather than an artefact of the three illustrative 4-bit
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 263 examples.

264 **C3. Regular interior and belief-simplex geometry.** We explicitly separate the
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 266 regular Walsh exponential family, where the usual Amari dual-flat structure and
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 268 Pythagorean identities apply, from the singular boundary orbit. The adversary's
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 270 key belief is treated as a separate full-support simplex, equipped with its own
 271
 272 Fisher–Rao metric.

273 **C4. CPA as Bayesian belief dynamics with pairwise leakage gaps.** Under
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 275 Gaussian leakage, each CPA trace is represented as a Bayesian update on the key-
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 belief simplex. The Hamming-weight variance $s/(4\sigma^2)$ is identified as a baseline

forced by bijectivity and uniform inputs. The quantities that distinguish key	277
hypotheses are the pairwise separations	278
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$\Delta(k, k^*) = \frac{1}{2^s} \sum_a (f(S(a \oplus k^*)) - f(S(a \oplus k)))^2,$	282
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which determine expected log-Bayes gains.	285
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C5. Reproducible diagnostics on lightweight S-boxes. For PRESENT, GIFT,	287
and SKINNY, we compute Walsh spectra, boundary covariance matrices, local	288
spectral separation proxies, pairwise Hamming-weight leakage gaps, and the pro-	289
jected active-eigenspace curvature diagnostic. The examples are used to separate	290
invariants forced by bijectivity or by the shared Walsh amplitude distribution	291
from quantities that can vary across S-boxes. The pairwise-gap tables identify the	292
hard key differences under the assumed CPA model, which is the security-relevant	293
information hidden by scalar amplitude-only summaries.	294
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1.5 Scope	302
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The paper focuses on isolated bijective S-boxes under uniform inputs and on an additive	304
Gaussian leakage model for CPA belief updates. It does not address complete multi-	305
round cipher security, masking, higher-order leakage, non-Gaussian leakage, profiling	306
attacks, or implementation-specific trace effects. Those topics require additional models	307
beyond the boundary distribution and belief-update framework developed here.	308
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2 Background	315
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2.1 S-boxes and the Walsh–Hadamard transform	318
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An <i>s</i> -bit <i>S</i> -box is a bijection $S : \{0, 1\}^s \rightarrow \{0, 1\}^s$. The <i>i</i> -th output bit is a Boolean	320
function $S_i : \{0, 1\}^s \rightarrow \{0, 1\}$.	321
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Definition 2.1 (Walsh–Hadamard Transform) The Walsh–Hadamard transform (WHT) of S at $(\alpha, \beta) \in \{0, 1\}^s \times \{0, 1\}^s$ is:

$$\widehat{S}(\alpha, \beta) = \sum_{a \in \{0, 1\}^s} (-1)^{\alpha \cdot a \oplus \beta \cdot S(a)} \in \{-2^s, \dots, +2^s\}, \quad 2 \mid \widehat{S}(\alpha, \beta). \quad (1)$$

By convention $\widehat{S}(0, 0) = 2^s$.

The WHT encodes the two fundamental cryptographic properties of S :

- **Nonlinearity:** $\text{NL}(S) = 2^{s-1} - \frac{1}{2} \max_{(\alpha, \beta) \neq (0, 0)} |\widehat{S}(\alpha, \beta)|$.
- **Differential uniformity:** $\delta(S) = \max_{\Delta_{\text{in}} \neq 0, \Delta_{\text{out}}} |\{a : S(a) \oplus S(a \oplus \Delta_{\text{in}}) = \Delta_{\text{out}}\}|$.

The *Linear Approximation Table* (LAT) entry is $\text{LAT}_S(\alpha, \beta) = \widehat{S}(\alpha, \beta)/2$. For any S-box, Parseval’s identity gives:

$$\sum_{(\alpha, \beta) \in \{0, 1\}^s \times \{0, 1\}^s} \widehat{S}(\alpha, \beta)^2 = 2^{2s} \cdot 2^s = 2^{3s}. \quad (2)$$

Example 2.1 (PRESENT S-box) The PRESENT-80 S-box ($s = 4$) is the hexadecimal table $S = [\text{C}, 5, 6, \text{B}, 9, 0, \text{A}, \text{D}, 3, \text{E}, \text{F}, 8, 4, 7, 1, 2]$. Its nonlinearity is $\text{NL}(S) = 4$ and differential uniformity $\delta(S) = 4$.

2.1.1 Remark on 4-bit S-box classification

The 4-bit case is small enough that strong classification results are available. In particular, optimal 4-bit S-boxes have been studied up to the standard equivalence relations used for vectorial Boolean functions [6]. Consequently, the numerical examples in Section 7 should not be interpreted as a new classification of PRESENT, GIFT, or SKINNY. Their role is narrower: they provide reproducible side-channel diagnostics for widely used lightweight S-boxes and illustrate which information-geometric quantities are forced by bijectivity or by common Walsh spectra.

2.2 Information geometry: essentials

We collect the minimal background from Amari and Nagaoka [1] and Ay et al. [2] required in the sequel.

2.2.1 Statistical manifolds and the Fisher metric

Let $\mathcal{X}$ be a finite set with $|\mathcal{X}| = N$. The *probability simplex* is $\Delta_{N-1} = \{p \in \mathbb{R}^N : p_x \geq 0, \sum_x p_x = 1\}$. Its interior $\mathring{\Delta}_{N-1}$ is an $(N-1)$ -dimensional manifold.

Definition 2.2 (Fisher–Rao Metric) The *Fisher–Rao metric* on $\mathring{\Delta}_{N-1}$ at p is the positive definite bilinear form

$$g_p(u, v) = \sum_{x \in \mathcal{X}} \frac{u_x v_x}{p_x}, \quad u, v \in T_p \mathring{\Delta}_{N-1} = \left\{ w \in \mathbb{R}^N : \sum_x w_x = 0 \right\}. \quad (3)$$

By Chentsov’s theorem [3], g_F is the unique Riemannian metric on $\mathring{\Delta}_{N-1}$ (up to a positive constant) that is invariant under sufficient statistics, making it the canonical statistical distance.

2.2.2 Exponential families as flat submanifolds

Definition 2.3 (Exponential Family) Given functions $T_1, \dots, T_d : \mathcal{X} \rightarrow \mathbb{R}$ (the *sufficient statistics*), the *exponential family* $\mathcal{E}$ is:

$$p(x; \theta) = \exp\left(\sum_{i=1}^d \theta_i T_i(x) - \psi(\theta)\right), \quad \theta \in \Theta \subseteq \mathbb{R}^d, \quad (4)$$

where $\psi(\theta) = \log \sum_x \exp(\sum_i \theta_i T_i(x))$ is the log-partition function.

The natural parameters θ_i and the expectation parameters $\eta_i = \mathbb{E}_\theta[T_i(X)]$ are related by $\eta = \nabla \psi(\theta)$ (the bijection is guaranteed when the family is minimal and regular).

415 **Theorem 2.1** (Dually Flat Structure, [1]) *Every regular minimal exponential family $\mathcal{E}$ is a*
416 *dually flat submanifold of $\mathring{\Delta}_{N-1}$: the e -connection (exponential connection) is flat in natural*
417 *parameters θ , and the m -connection (mixture connection) is flat in expectation parameters η .*
418 *The generalised Pythagorean theorem holds: if $P_A, P_B, P_C \in \mathcal{E}$ and the e -geodesic from A to*
419 *B is g_F -orthogonal to the m -geodesic from B to C , then*

$$422 \quad D_{\text{KL}}(P_A \| P_C) = D_{\text{KL}}(P_A \| P_B) + D_{\text{KL}}(P_B \| P_C). \quad (5)$$

427 2.2.3 Mixture projection and Bayesian inference

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429 **Definition 2.4** (m -Projection) *Given $p \in \mathring{\Delta}_{N-1}$ and a closed m -flat submanifold $\mathcal{M} \subseteq \mathring{\Delta}_{N-1}$,*
430 *the m -projection of p onto $\mathcal{M}$ is:*

$$432 \quad \Pi_{\mathcal{M}}^{(m)}(p) = \arg \min_{q \in \mathcal{M}} D_{\text{KL}}(q \| p). \quad (6)$$

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437 The following classical result (see [1], Chapter 3) is central to Section 5:

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441 **Proposition 2.1** (Bayesian Update as m -Projection) *Let $p(x, y) = p(x)p(y|x)$ be a joint*
442 *distribution on $\mathcal{X} \times \mathcal{Y}$. After observing $y = y_0$, the posterior $p(x|y_0)$ is the m -projection of*
443 *the joint p onto the submanifold $\mathcal{M}_{y_0} = \{q \in \Delta : q(y) = 0 \text{ for } y \neq y_0\}$, restricted to the*
444 *x -marginal.*

450 2.2.4 The LDPC analogy: Ikeda-Tanaka-Amari (2004)

451
452 Ikeda, Tanaka, and Amari [4] develop a complete information-geometric framework for
453 LDPC and turbo codes. They model the joint distribution over a binary codeword
454 $x \in \{-1, +1\}^N$ as:

$$455 \quad p(x; \theta, v) = \exp\left(\theta \cdot x + \sum_r v_r c_r(x) - \psi\right), \quad (7)$$

where $c_r(x) = \prod_{i \in r} x_i$ are parity check functions and v_r the constraint weights. They show: (i) the e/m dual structure is exact; (ii) Belief Propagation decoding corresponds to iterated e/m -projections; (iii) curvature of the statistical manifold controls decoding convergence.

In the present paper, the parity functions $c_r(x)$ are replaced by the Walsh-Hadamard characters $\chi_{\alpha\beta}(a, b)$ of the S-box, and LDPC decoding is replaced by CPA key recovery. Table 2 summarises the correspondence.

Table 2 Structural analogy between the LDPC framework of Ikeda-Tanaka-Amari (2004) and the S-box framework of this paper.

Ikeda-Tanaka-Amari (LDPC)	This paper (S-box)
Codeword $x \in \{-1, +1\}^N$	Input-output pair $(a, b) \in \{0, 1\}^{2s}$
Parity check $c_r(x) = \prod_{i \in r} x_i$	Walsh character $\chi_{\alpha\beta}(a, b) = (-1)^{\alpha \cdot a \oplus \beta \cdot b}$
Channel log-likelihood θ_i	Key log-likelihood ratio $\log(q_k/q_{k'})$
Parity constraint weight v_r	Walsh spectrum value $\hat{S}(\alpha, \beta)$
Belief Propagation decoding	CPA key recovery
Decoding convergence rate	Per-trace signal information $I_S(\sigma)$

3 The S-box Statistical Model

3.1 Key-induced boundary distributions

Fix a bijective S-box $S : \{0, 1\}^s \rightarrow \{0, 1\}^s$ and let A be uniformly distributed over $\{0, 1\}^s$. Under key hypothesis $k \in \{0, 1\}^s$, the observed input-output pair is $(A, S(A \oplus k))$. This induces the following distribution on $\mathcal{X} = \{0, 1\}^s \times \{0, 1\}^s$.

Definition 3.1 (Key-induced distribution) For $k \in \{0, 1\}^s$, define

$$P_k(a, b) = 2^{-s} \mathbf{1}[b = S(a \oplus k)]. \quad (8)$$

The support of P_k is the graph $G_k = \{(a, S(a \oplus k)) : a \in \{0, 1\}^s\}$.

Each P_k is a boundary point of the simplex $\Delta_{2^{2s}-1}$, because it assigns zero mass to $2^{2s} - 2^s$ outcomes. This support deficiency is central: ordinary Fisher–Rao tensors of the full simplex are defined on the interior, not directly at P_k . We therefore treat P_k as a singular endpoint of an interior regularization when differential-geometric operations are needed.

3.2 Walsh representation and key translation

Definition 3.2 (Walsh coefficient of P_k) For $(\alpha, \beta) \in \{0, 1\}^s \times \{0, 1\}^s$, set

$$\hat{P}_k(\alpha, \beta) = \sum_{(a,b) \in \mathcal{X}} \chi_{\alpha\beta}(a, b) P_k(a, b) = 2^{-s} \sum_{a \in \{0,1\}^s} (-1)^{\alpha \cdot a \oplus \beta \cdot S(a \oplus k)}. \quad (9)$$

Lemma 3.1 (Walsh key-translation rule) For every $k \in \{0, 1\}^s$ and every $(\alpha, \beta) \in \{0, 1\}^s \times \{0, 1\}^s$,

$$\hat{P}_k(\alpha, \beta) = (-1)^{\alpha \cdot k} \hat{P}_0(\alpha, \beta) = 2^{-s} (-1)^{\alpha \cdot k} \hat{S}(\alpha, \beta). \quad (10)$$

Proof Using $a' = a \oplus k$,

$$\hat{P}_k(\alpha, \beta) = 2^{-s} \sum_{a'} (-1)^{\alpha \cdot (a' \oplus k) \oplus \beta \cdot S(a')} = 2^{-s} (-1)^{\alpha \cdot k} \sum_{a'} (-1)^{\alpha \cdot a' \oplus \beta \cdot S(a')},$$

which is exactly the claimed identity. $\square$

Remark 3.1 (Walsh orbit) The key does not change the amplitude $|\hat{S}(\alpha, \beta)|$; it only changes the sign according to the character $\alpha \mapsto (-1)^{\alpha \cdot k}$. Thus $\{P_k : k \in \{0, 1\}^s\}$ is best described as a finite Walsh-phase orbit, not as a smooth manifold. The rule is a structural translation identity; the information-geometric content in the sequel comes from the covariance limit, active subspace, and belief-update geometry built on top of this identity.

3.3 The regular Walsh exponential family

Definition 3.3 (Walsh exponential family) Let $T_{\alpha\beta} = \chi_{\alpha\beta}$ for $(\alpha, \beta) \neq (0, 0)$. The Walsh exponential family on $\mathcal{X}$ is

$$p_{\theta}(a, b) = \exp \left(\sum_{(\alpha, \beta) \neq (0, 0)} \theta_{\alpha\beta} T_{\alpha\beta}(a, b) - \psi(\theta) \right), \quad \theta \in \mathbb{R}^{2^{2s}-1}. \quad (11)$$

Because the nonconstant Walsh characters together with the constant character form an orthogonal basis of all real functions on $\mathcal{X}$, this is simply the full regular exponential family over $\mathcal{X}$ written in Walsh coordinates. Its image is the interior of the probability simplex. The singular distributions P_k belong to the closure, not to the regular parameter space.

Proposition 3.1 (Boundary expectation coordinates) For each k and $(\alpha, \beta) \neq (0, 0)$,

$$\eta_{\alpha\beta}^{(k)} := \mathbb{E}_{P_k}[T_{\alpha\beta}] = (-1)^{\alpha \cdot k} 2^{-s} \widehat{S}(\alpha, \beta). \quad (12)$$

The corresponding natural parameters are not finite; they are obtained only as limits of full-support approximations approaching P_k .

Proof The first equality is the definition of expectation coordinates and the second is Lemma 3.1. Since P_k has zeros on $\mathcal{X}$, no finite vector θ in (11) can represent it. $\square$

3.4 Interior regularization and boundary covariance form

Let U denote the uniform distribution on $\mathcal{X}$. For $\varepsilon \in (0, 1)$ define

$$P_k^{\varepsilon} = (1 - \varepsilon)P_k + \varepsilon U. \quad (13)$$

Then P_k^{ε} has full support and belongs to the regular simplex interior. Fisher–Rao calculations are unambiguous at P_k^{ε} . The finite covariance limit as $\varepsilon \downarrow 0$ is the following.

599 **Definition 3.4** (Boundary Fisher covariance form) The boundary covariance form associated
600 with key k is
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$$602 \quad \mathcal{I}_{ij}^\partial(k) = \text{Cov}_{P_k}(T_i, T_j), \quad i, j \in \{0, 1\}^s \times \{0, 1\}^s \setminus \{(0, 0)\}. \quad (14)$$

603
604 It is positive semidefinite and generally singular. We do not call it a non-degenerate Fisher
605 metric at P_k .
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609 **Proposition 3.2** (Closed form of the boundary covariance) For $i = (\alpha, \beta)$ and $j = (\alpha', \beta')$,
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$$611 \quad \begin{aligned} 612 \quad \mathcal{I}_{ij}^\partial(k) &= \text{Cov}_{P_k}(T_{\alpha\beta}, T_{\alpha'\beta'}) \\ 613 \quad &= (-1)^{(\alpha \oplus \alpha') \cdot k} \left(2^{-s} \widehat{S}(\alpha \oplus \alpha', \beta \oplus \beta') - 2^{-2s} \widehat{S}(\alpha, \beta) \widehat{S}(\alpha', \beta') \right). \end{aligned} \quad (15)$$

614
615 In particular,
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$$617 \quad \mathcal{I}_{(\alpha\beta), (\alpha\beta)}^\partial(k) = 1 - \left(2^{-s} \widehat{S}(\alpha, \beta) \right)^2. \quad (16)$$

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622 *Proof* Since $T_i T_j = T_{i \oplus j}$,
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$$624 \quad \mathbb{E}_{P_k}[T_i T_j] = \widehat{P}_k(\alpha \oplus \alpha', \beta \oplus \beta') = (-1)^{(\alpha \oplus \alpha') \cdot k} 2^{-s} \widehat{S}(\alpha \oplus \alpha', \beta \oplus \beta').$$

625
626 Subtracting $\mathbb{E}_{P_k}[T_i] \mathbb{E}_{P_k}[T_j]$ and applying Lemma 3.1 gives (15). The diagonal follows from
627 $T_i^2 = 1$. □
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630
631 **Lemma 3.2** (Key covariance isometry) Let D_k be the diagonal sign matrix $(D_k)_{(\alpha\beta), (\alpha\beta)} =$
632 $(-1)^{\alpha \cdot k}$. Then
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$$634 \quad \mathcal{I}^\partial(k) = D_k \mathcal{I}^\partial(0) D_k. \quad (17)$$

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636 Thus every spectral invariant of the boundary covariance form is independent of the true key.
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642 *Proof* Equation (15) factors exactly into the sign contribution $(-1)^{\alpha \cdot k} (-1)^{\alpha' \cdot k}$ times the
643 $k = 0$ covariance entry. □
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Remark 3.2 (What is metric and what is not) For $\varepsilon > 0$, the Fisher–Rao metric at P_k^ε is an ordinary Riemannian metric. At $\varepsilon = 0$, only the finite covariance form (14) is used. Distances or curvatures involving the exact P_k are therefore either (i) computed on the belief simplex, (ii) computed for P_k^ε and then studied as limits, or (iii) explicitly described as first-order/local covariance approximations.

3.5 First-order spectral separation of key hypotheses

The exact distributions P_k and $P_{k'}$ have disjoint supports when $k \neq k'$, so their KL divergence is infinite. Nevertheless, the Walsh coordinate displacement has a simple closed form:

$$\eta_{\alpha\beta}^{(k')} - \eta_{\alpha\beta}^{(k)} = \left((-1)^{\alpha \cdot k'} - (-1)^{\alpha \cdot k} \right) 2^{-s} \widehat{S}(\alpha, \beta). \quad (18)$$

Consequently, only Walsh modes satisfying $\alpha \cdot (k \oplus k') = 1$ contribute to the spectral separation. A diagonal local approximation based on the boundary covariance form is

$$d_{\text{loc}}^2(k, k') = \sum_{\substack{(\alpha, \beta) \neq (0, 0): \\ \alpha \cdot (k \oplus k') = 1}} \frac{4 \cdot 2^{-2s} \widehat{S}(\alpha, \beta)^2}{1 - 2^{-2s} \widehat{S}(\alpha, \beta)^2}, \quad (19)$$

provided the denominator is nonzero. This quantity is a local spectral proxy, not the exact Fisher–Rao geodesic distance between the singular distributions.

4 Dual Structure, Boundary Limits, and Belief Geometry

4.1 Regular dual flatness of the interior family

Theorem 4.1 (Regular dual flatness) *The Walsh exponential family (11) is a regular minimal exponential family on $\mathcal{X}$. Its interior carries the standard dually flat information-geometric structure: natural coordinates θ are e -affine, expectation coordinates η are m -affine, the log-partition function ψ is the e -potential, and its Legendre dual is $\varphi(\eta) = -H(p_\eta)$.*

691 *Proof* The natural parameter space is the open set $\mathbb{R}^{2^s-1}$. The sufficient statistics are
692 bounded, so the partition function is finite and smooth for all θ . Minimality follows from the
693 orthogonality and linear independence of the nonconstant Walsh characters together with the
694 constant character. The standard theorem for regular minimal exponential families then gives
695 dual flatness and the Legendre correspondence [1, 2]. $\square$

698
699
700
701 *Remark 4.1* (Boundary discipline) Theorem 4.1 applies to full-support distributions only. The
702 exact key distributions P_k lie in the closure. Statements about geodesics, Christoffel symbols,
703 Bregman identities, or curvature at P_k are therefore meaningful only after regularization or
704 after being rephrased as finite boundary covariance statements.

710 4.2 KL divergence between exact key distributions

711
712 **Proposition 4.1** (Support separation) *If S is bijective and $k \neq k'$, then*

$$714 \quad \text{supp}(P_k) \cap \text{supp}(P_{k'}) = \emptyset, \quad D_{\text{KL}}(P_k \| P_{k'}) = +\infty. \quad (20)$$

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723 *Proof* If (a, b) belongs to both supports, then $S(a \oplus k) = b = S(a \oplus k')$. Since S is injective,
724 $a \oplus k = a \oplus k'$, hence $k = k'$, a contradiction. Thus $P_{k'}$ assigns zero probability to every point
725 in the support of P_k when $k \neq k'$, which implies infinite KL divergence. $\square$

727
728
729
730 *Remark 4.2* (No finite Bregman identity at singular endpoints) The usual equality between
731 KL divergence and the Bregman divergence of the dual potential is an interior statement. For
732 P_k and $P_{k'}$ with disjoint supports, one may study $D_{\text{KL}}(P_k^\varepsilon \| P_{k'}^\varepsilon)$ and then let $\varepsilon \downarrow 0$, but the
733 limiting divergence is infinite. We therefore do not use finite Pythagorean decompositions
734 directly between distinct exact key distributions.

4.3 Mixtures induced by key beliefs

Let $Q = \Delta_{2^s-1}^\circ$ be the interior of the simplex over keys. A belief $q \in Q$ induces a distribution on input-output pairs

$$\bar{P}_q(a, b) = \sum_{k \in \{0,1\}^s} q(k) P_k(a, b) = 2^{-s} \sum_{k \in \{0,1\}^s} q(k) \mathbf{1}[b = S(a \oplus k)]. \quad (21)$$

If q has full support, then $\bar{P}_q$ has full support on $\mathcal{X}$.

Proposition 4.2 (Walsh representation of belief mixtures) *Let*

$$\hat{q}(\alpha) = \sum_{k \in \{0,1\}^s} q(k) (-1)^{\alpha \cdot k}.$$

Then

$$\bar{P}_q(a, b) = 2^{-2s} \sum_{\alpha, \beta \in \{0,1\}^s} \hat{q}(\alpha) \hat{S}(\alpha, \beta) (-1)^{\alpha \cdot a \oplus \beta \cdot b}. \quad (22)$$

In particular, the uniform belief induces the uniform distribution on $\mathcal{X}$.

Proof Apply Walsh inversion to the indicator $\mathbf{1}[b = S(a \oplus k)]$ and sum over k with weights $q(k)$. For uniform q , $\hat{q}(\alpha) = \mathbf{1}[\alpha = 0]$, and $\hat{S}(0, 0) = 2^s$ while $\hat{S}(0, \beta) = 0$ for $\beta \neq 0$ because S is bijective. $\square$

4.4 Fisher–Rao geometry of the belief simplex

Definition 4.1 (Belief simplex metric) For $q \in Q$ and tangent vectors u, v with $\sum_k u_k = \sum_k v_k = 0$, the Fisher–Rao metric is

$$g_q^{(Q)}(u, v) = \sum_{k \in \{0,1\}^s} \frac{u_k v_k}{q(k)}. \quad (23)$$

783 Under the square-root map $q \mapsto (\sqrt{q(k)})_k$, this metric is the round metric on a
 784 sphere of radius 2. Therefore
 785

$$786 \quad d_F(q, q') = 2 \arccos \left(\sum_{k \in \{0,1\}^s} \sqrt{q(k)q'(k)} \right). \quad (24)$$

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 791 For distinct point-mass limits, $d_F(\delta_k, \delta_{k'}) = \pi$. This is a statement about the belief
 792 simplex, not about geodesics between the singular input-output distributions P_k and
 793 $P_{k'}$.
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810 4.5 Pythagorean identities

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 812 The generalized Pythagorean theorem is used in the following restricted sense. For
 813 full-support distributions in the regular Walsh exponential family, the standard identity
 814
 815

$$816 \quad D_{\text{KL}}(P_A \| P_C) = D_{\text{KL}}(P_A \| P_B) + D_{\text{KL}}(P_B \| P_C) \quad (25)$$

817
 818
 819
 820 holds whenever the e-geodesic from A to B is orthogonal to the m-geodesic from B to
 821 C . In the CPA setting, the relevant full-support object is the posterior belief $q_n \in Q$
 822 as long as the prior has full support and the Gaussian likelihood is strictly positive.
 823 This is why projection statements in Section 5 are made on Q or on regularized joint
 824 distributions, not directly on disjoint-support P_k endpoints.
 825
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 827
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5 Correlation Power Analysis as Bayesian Projection Dynamics

5.1 Leakage model

We consider the standard non-profiled CPA setting. The adversary knows S , observes plaintexts $a_1, \dots, a_n$ uniformly drawn from $\{0, 1\}^s$, and obtains leakage measurements

$$W_i = f(S(a_i \oplus k^*)) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \quad (26)$$

where k^* is the true key and $f : \{0, 1\}^s \rightarrow \mathbb{R}$ is the leakage model. In the numerical experiments we take $f = \text{HW}$.

5.2 Bayesian update as a mixture projection

Theorem 5.1 (CPA posterior update as m -projection) *Let $q_{i-1} \in \Delta_{2^s-1}^{\circ}$ be the prior belief over keys before observing (a_i, W_i) . The posterior*

$$q_i(k) = \frac{q_{i-1}(k) p(W_i | a_i, k)}{\sum_{h \in \{0, 1\}^s} q_{i-1}(h) p(W_i | a_i, h)} \quad (27)$$

is the key marginal of the regular conditional distribution obtained from the joint prior-likelihood model. Equivalently, it can be obtained as the limit of m -projections after conditioning on shrinking measurement bins around the observed continuous value.

Proof The joint distribution before conditioning is $q_{i-1}(k)p(a_i)p(W_i | a_i, k)$. Since the Gaussian observation is continuous, the event $W_i = w_i$ has probability zero and conditioning is understood through a regular conditional density, or equivalently as the limit obtained by conditioning on $W_i \in [w_i - \epsilon, w_i + \epsilon]$ and letting $\epsilon \downarrow 0$. The limiting conditional law has key marginal exactly (27). $\square$

The exact S-box distributions P_k lie on the boundary of the simplex, but the belief trajectory $q_0, q_1, \dots, q_n$ stays in the interior of the belief simplex as long as the prior

875 has full support and the Gaussian likelihood is positive. Hence CPA can be studied as
 876 a regular trajectory on $(Q, g_F^{(Q)})$.
 877

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880 5.3 Per-trace signal information

881

882 **Definition 5.1** (Per-trace signal information) For a leakage model f and noise variance σ^2 ,
 883 define
 884

$$885 I_S(f, \sigma) = \sigma^{-2} \text{Var}_{a \sim \text{Unif}(\{0,1\}^s)} [f(S(a \oplus k^*))]. \quad (28)$$

886

887 When S is bijective, this quantity does not depend on k^* .
 888

889

890 **Proposition 5.1** (Hamming-weight specialization) For $f = \text{HW}$ and a bijective S ,

891

$$892 I_S(\text{HW}, \sigma) = \frac{s}{4\sigma^2}. \quad (29)$$

893

894 In particular, for 4-bit bijective S -boxes, $I_S(\text{HW}, \sigma) = 1/\sigma^2$.
 895

896

897

898 *Proof* Bijectivity of S together with the uniformity of $a \oplus k^*$ implies that $S(a \oplus k^*)$ is uniform
 899 over $\{0, 1\}^s$. The Hamming weight of a uniform s -bit word follows a binomial distribution
 900 Binomial($s, 1/2$), whose variance is $s/4$. □
 901

902

903

904 5.4 Pairwise log-likelihood increments

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906 For a competing key $k \neq k^*$, the per-trace squared separation is
 907

908

909

$$910 \Delta(k, k^*) = \frac{1}{2^s} \sum_{a \in \{0,1\}^s} (f(S(a \oplus k^*)) - f(S(a \oplus k)))^2. \quad (30)$$

911

912

913

914 **Proposition 5.2** (Expected pairwise log-Bayes gain) Under the Gaussian leakage model,

915

$$916 \mathbb{E} \left[\log \frac{p(W | A, k^*)}{p(W | A, k)} \right] = \frac{\Delta(k, k^*)}{2\sigma^2}, \quad (31)$$

917

918 where the expectation is over A (uniform) and W conditioned on A and k^* .
 919

920

Proof For a fixed plaintext a , the two likelihoods are Gaussian with the same variance σ^2 and means $\mu_*(a) = f(S(a \oplus k^*))$, $\mu_k(a) = f(S(a \oplus k))$. The Kullback-Leibler divergence between two univariate Gaussians with equal variance is $(\mu_*(a) - \mu_k(a))^2 / (2\sigma^2)$. Averaging over a gives the result. $\square$

The worst-case pairwise gap

$$\Gamma_S(f, \sigma) = \min_{k \neq k^*} \frac{\Delta(k, k^*)}{2\sigma^2} \quad (32)$$

provides a rigorous likelihood-separation measure, distinct from $I_S(f, \sigma)$ in general.

5.5 Model-based sample-complexity estimate

Starting from a uniform prior, the initial key uncertainty equals $s \log 2$ nats. A first-order information estimate then yields

$$n_{\text{est}} \approx \frac{s \log 2}{I_S(f, \sigma)}. \quad (33)$$

For 4-bit Hamming-weight leakage, $n_{\text{est}} \approx 4 \log 2 \sigma^2$. This is a heuristic leakage-rate estimate. A more rigorous bound follows from the pairwise gap $\Gamma_S(f, \sigma)$: the expected log-posterior odds against any fixed wrong key grow at rate at least $\Gamma_S(f, \sigma)$. Thus reaching an expected odds margin τ against the hardest wrong key requires roughly

$$n_{\text{gap}}(\tau) \approx \frac{\tau}{\Gamma_S(f, \sigma)} = \frac{2\sigma^2\tau}{\min_{k \neq k^*} \Delta(k, k^*)}. \quad (34)$$

This estimate is model-based rather than an attack-success theorem, but it is operational: two S-boxes with the same scalar I_S can lead to different worst-case posterior-odds growth if their minimum pairwise gaps differ.

967 **6 Active-Eigenspace Curvature: Computation and**
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969 **Status**
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972 **6.1 What curvature is being computed**
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974 The regular Walsh exponential family is dually flat with respect to the e- and m-
975 connections, but its Levi-Civita curvature is not automatically zero. At exact key
976 distributions P_k , however, the ordinary Riemannian tensor of the full-support family
977 is not directly defined because P_k lies on the boundary. This section therefore uses the
978 following restricted object.
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983
984 **Definition 6.1** (Active covariance eigenspace) Let $\mathcal{I}^\partial(k)$ be the boundary covariance matrix
985 in Definition 3.4. Its active eigenspace is the span of the eigenvectors with strictly positive
986 eigenvalues. Curvature computations below are performed after projecting the covariance and
987 third cumulant tensors to this active eigenspace.
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992 The resulting values are curvature diagnostics of the limiting regularized geometry
993 obtained from the active covariance eigenspace. The finite orbit $\{P_k\}$ itself is discrete
994 and is used here through its induced boundary covariance and cumulant tensors.
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999 **6.2 Universal rank and eigenvalues of the boundary covariance**
1000
1001 **Theorem 6.1** (Rank and active eigenvalues) *For any bijection $S : \{0, 1\}^s \rightarrow \{0, 1\}^s$, the*
1002 *boundary covariance matrix $\mathcal{I}^\partial(k)$ has rank $2^s - 1$. Its nonzero eigenvalues are all 2^s when the*
1003 *feature coordinates are the unnormalised nonconstant Walsh characters used in Definition 3.3.*
1004
1005
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1008
1009 *Proof* Let $N = 2^s$ and index rows by $a \in \{0, 1\}^s$. Let X be the $N \times (2^{2s} - 1)$ feature
1010 matrix with entries $X_{a,(\alpha,\beta)} = (-1)^{\alpha \cdot a \oplus \beta \cdot S(a \oplus k)}$ for $(\alpha, \beta) \neq (0, 0)$. The boundary covariance
1011 is $N^{-1} X^\top H X$, where $H = I_N - N^{-1} \mathbf{1}\mathbf{1}^\top$ centers over the support graph. The nonzero
1012

eigenvalues of $N^{-1}X^\top HX$ equal those of $N^{-1}HXX^\top H$. For $a, a' \in \{0, 1\}^s$,

$$(XX^\top)_{a,a'} = \sum_{(\alpha,\beta) \neq (0,0)} (-1)^{\alpha \cdot (a \oplus a') \oplus \beta \cdot (S(a \oplus k) \oplus S(a' \oplus k))} = N^2 \mathbf{1}[a = a'] - 1,$$

because S is injective. Hence $N^{-1}HXX^\top H = N^{-1}H(N^2I_N - \mathbf{1}\mathbf{1}^\top)H = NH$. The matrix H has rank $N - 1$ and eigenvalue 1 on the centered subspace. Therefore the active eigenvalues are all $N = 2^s$. $\square$

6.3 Third cumulants and curvature protocol

Let

$$C_{ijk}^\partial(k) = \mathbb{E}_{P_k} [(T_i - \eta_i)(T_j - \eta_j)(T_k - \eta_k)] \quad (35)$$

be the boundary third cumulant tensor. Since products of Walsh characters are again Walsh characters, every entry can be computed exactly from the WHT of S and the key-translation rule. If U is an orthonormal basis of the active eigenspace and $\lambda_a = 2^s$ are the active covariance eigenvalues, the projected tensor is

$$\tilde{C}_{abc} = \sum_{i,j,\ell} U_{ia} U_{jb} U_{\ell c} C_{ij\ell}^\partial. \quad (36)$$

The Levi-Civita curvature diagnostic used in the experiments is then computed with the standard dually-flat formula

$$R_{abab} = \frac{1}{4} \sum_c \frac{\tilde{C}_{aac} \tilde{C}_{bbc} - \tilde{C}_{abc}^2}{\lambda_c}, \quad K_{ab} = \frac{R_{abab}}{\lambda_a \lambda_b}, \quad a < b. \quad (37)$$

Both verification scripts implement Equation (37) directly.

1059 6.4 Empirical constant-curvature finding

1060

1061 **Proposition 6.1** (Reproducible 4-bit curvature observation) *For the PRESENT, GIFT, and*
1062 *SKINNY 4-bit S-boxes, the projected active-eigenspace curvature diagnostic (37) gives*
1063

$$1064 K_{ab} = -\frac{1}{4} \quad (38)$$

1066

1067 *for all $\binom{15}{2} = 105$ active coordinate planes, up to numerical precision in the provided scripts.*

1068

1069

1070

1071 *Verification protocol* The computation is reproducible from the repository scripts. The scripts
1072 build the WHT table, construct $\mathcal{I}^\theta$, diagonalize its active 15-dimensional eigenspace, project
1073 the third cumulant tensor, and evaluate (37). The reported value is stable across the three
1074 tested S-boxes and an additional random 4-bit bijection in `verify_curvature.py`.

1075 These reproducible computations suggest the following conjecture: for bijective S-boxes,
1076 the active Fisher subspace associated with the Walsh boundary covariance form has constant
1077 sectional curvature $K = -1/4$, under the curvature convention and projection procedure used
1078 in this work. A symbolic proof, or a characterization of the precise hypotheses under which
1079 the conjecture holds, is left as an open problem. $\square$

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1085

1086 6.5 Relation to the belief simplex

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1088 The belief simplex $Q = \Delta_{2^s-1}^\circ$ with Fisher–Rao metric is isometric, via the square-root
1089 embedding, to a sphere of radius 2 and hence has constant sectional curvature $+1/4$.

1090 The observed value $-1/4$ in the active boundary covariance diagnostic suggests a sign-
1091 dual phenomenon between the posterior belief geometry and the limiting keyed S-box
1092 covariance geometry, which is recorded here as part of the open geometric structure of
1093 the model.

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7 Reproducible Security Diagnostics

The computations in this section are not intended to classify 4-bit S-boxes or to
benchmark full cipher attacks. They serve two narrower purposes. First, they verify

the exact boundary covariance and active-eigenspace formulas on standard lightweight
S-boxes. Second, they separate quantities that are structural invariants from quantities
that can influence CPA discrimination between key hypotheses.

7.1 Target S-boxes

We evaluate the 4-bit bijective S-boxes used in PRESENT, GIFT, and SKINNY [13–15]:

$$S_{\text{PRESENT}} = [C, 5, 6, B, 9, 0, A, D, 3, E, F, 8, 4, 7, 1, 2],$$

$$S_{\text{GIFT}} = [1, A, 4, C, 6, F, 3, 9, 2, D, B, 7, 5, 0, 8, E],$$

$$S_{\text{SKINNY}} = [C, 6, 9, 0, 1, A, 2, B, 3, 8, 5, D, 4, E, 7, F].$$

These examples are useful because they are widely used in lightweight cryptography
and because exhaustive enumeration over all inputs, masks, and key hypotheses is
possible.

7.2 Walsh spectra and boundary covariance

For each S-box we compute the full WHT table

$$\hat{S}(\alpha, \beta) = \sum_{a \in \{0,1\}^4} (-1)^{\alpha \cdot a \oplus \beta \cdot S(a)}.$$

The three tested S-boxes have the same absolute Walsh-coefficient distribution over
nonzero masks:

$$\#|\hat{S}| = 0 : 123, \quad \#|\hat{S}| = 4 : 96, \quad \#|\hat{S}| = 8 : 36.$$

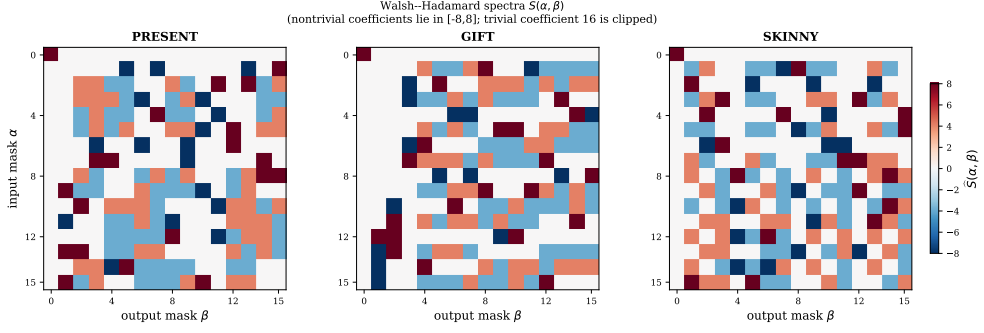


Fig. 2 Walsh spectra. The plotted Walsh tables show the same amplitude profile for PRESENT, GIFT, and SKINNY, which explains why purely amplitude-based diagnostics coincide for these examples.

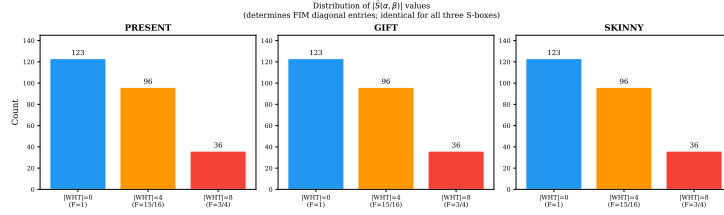


Fig. 3 Walsh amplitude distribution. The histogram records the shared counts of absolute Walsh coefficients over nonzero masks for the three tested S-boxes.

Table 3 Boundary covariance structure for the tested 4-bit bijections.

S-box	Matrix size	Rank	Active eigenvalues
PRESENT	255×255	15	16 repeated 15 times
GIFT	255×255	15	16 repeated 15 times
SKINNY	255×255	15	16 repeated 15 times

Therefore any diagnostic depending only on the Walsh amplitudes must coincide for these three examples. This is an expected consequence of the spectral data, not a new distinguishing property.

The boundary covariance matrix is built from Proposition 3.2. For $s = 4$ it is a 255×255 positive semidefinite matrix. Consistent with Theorem 6.1, all three matrices have rank 15 and active eigenvalues equal to 16 up to numerical precision.

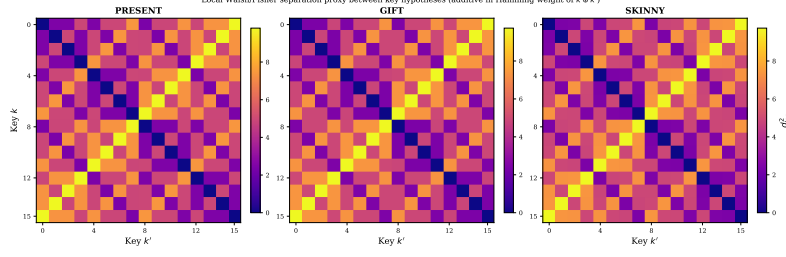


Fig. 4 Local key-separation proxy. The matrix visualises the covariance-weighted Walsh displacement between key hypotheses and should be read as a local diagnostic, not as an exact distance between singular laws.

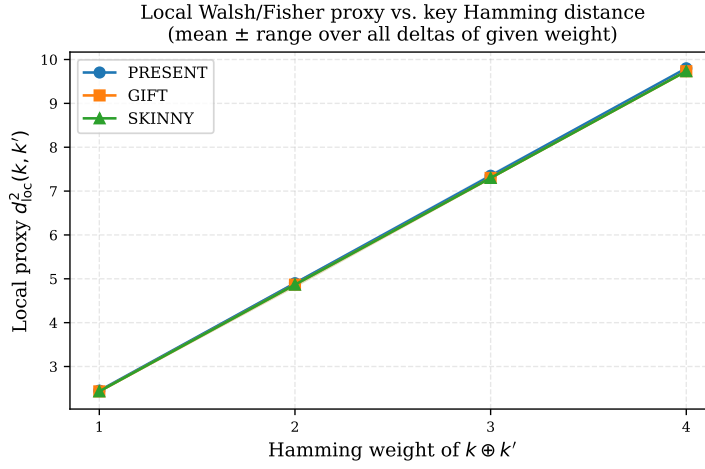


Fig. 5 Local proxy versus key-difference Hamming weight. The plot aggregates the same covariance-weighted Walsh displacement by the Hamming weight of $k \oplus k'$; it is not an exact Fisher–Rao distance between the singular distributions.

7.3 Local spectral separation

Using (19), we compute local diagonal Walsh/Fisher separation proxies between key hypotheses. This value is not the exact Fisher–Rao distance between singular distributions. It is a local covariance-weighted coordinate displacement. The key-translation rule implies that the value depends only on the key difference $k \oplus k'$.

Table 4 Interpretation of the key-separation quantities.

Quantity	Status
$D_{\text{KL}}(P_k \ P_{k'})$ for $k \neq k'$	Infinite, because supports are disjoint
$d_F(\delta_k, \delta_{k'})$ on the belief simplex	Exactly π
$d_{\text{loc}}^2(k, k')$ in (19)	Local Walsh/Fisher covariance proxy

Table 5 Pairwise Hamming-weight leakage gaps over nonzero key differences.

S-box	$\min_{\delta \neq 0} \Delta(\delta)$	$\max_{\delta \neq 0} \Delta(\delta)$	$\frac{1}{15} \sum_{\delta \neq 0} \Delta(\delta)$
PRESENT	1	7/2	32/15
GIFT	5/4	7/2	32/15
SKINNY	1	3	32/15

7.4 CPA leakage gaps

For bijective 4-bit S-boxes and uniform inputs, the scalar Hamming-weight variance in Proposition 5.1 is always

$$I_S(\text{HW}, \sigma) = \frac{1}{\sigma^2}.$$

This scalar is a baseline leakage descriptor and cannot distinguish PRESENT, GIFT, and SKINNY. Pairwise gaps

$$\Delta(\delta) = \frac{1}{16} \sum_{a \in \{0,1\}^4} (\text{HW}(S(a)) - \text{HW}(S(a \oplus \delta)))^2, \quad \delta \neq 0,$$

are more informative for CPA because the expected log-Bayes gain against a wrong key with difference δ is $\Delta(\delta)/(2\sigma^2)$.

Table 5 shows why the scalar variance descriptor should not be overinterpreted: all three S-boxes have the same average pairwise gap, but the worst and best key-difference cases vary. These pairwise gaps are the quantities that enter expected posterior-odds growth in the Gaussian leakage model.

Table 6 Operational CPA interpretation of the pairwise gaps. The listed deltas are nonzero key differences $\delta = k \oplus k^*$ in hexadecimal.

S-box	Hardest min Δ	gaps Easiest max Δ	gaps Gap multiset over $\delta \neq 0$
PRESENT	1 at $\{A, E\}$	7/2 at $\{6\}$	1^2 ; $(5/4)^1$; $(3/2)^2$; $(7/4)^3$; $(5/2)^2$; $(11/4)^1$; 3^2 ; $(13/4)^1$; $(7/2)^1$
GIFT	$5/4$ $\{C, E, F\}$	at 7/2 at $\{7\}$	$(5/4)^3$; $(3/2)^2$; 2^2 ; $(9/4)^3$; $(5/2)^2$; 3^2 ; $(7/2)^1$
SKINNY	1 at $\{5\}$	3 at $\{B\}$	1^1 ; $(5/4)^1$; $(3/2)^1$; $(7/4)^1$; 2^3 ; $(9/4)^2$; $(5/2)^3$; $(11/4)^2$; 3^1

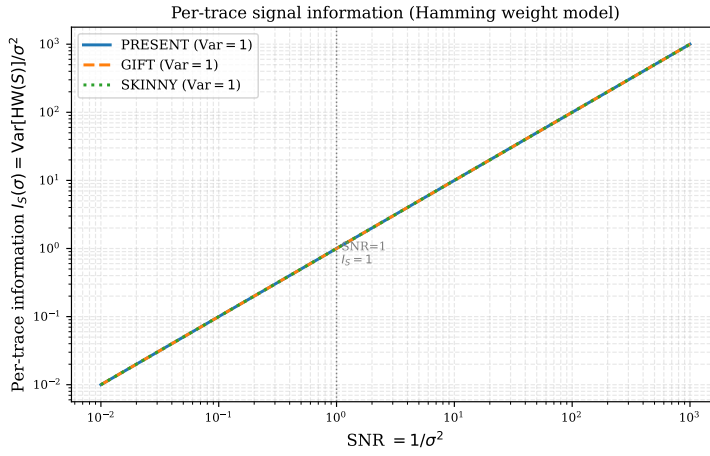


Fig. 6 Per-trace leakage baseline. The scalar Hamming-weight variance is the same for all bijective 4-bit S-boxes under uniform inputs, so CPA discrimination must be interpreted through pairwise leakage gaps rather than this baseline alone.

Table 6 is the part of the experiment that is most directly relevant to CPA. PRESENT and SKINNY have a smaller hardest-key gap than GIFT under this leakage model, so the model-based estimate (34) predicts a larger number of traces to obtain the same expected odds margin against the hardest wrong key. This observation does not imply a full-cipher security ordering, but it shows why the proposed separation between scalar invariants and pairwise discrimination structure is necessary.

Table 7 Projected active-eigenspace curvature diagnostic for the tested 4-bit S-boxes.

S-box	Active dimension	Planes	Min K	Max K
PRESENT	15	105	−0.25	−0.25
GIFT	15	105	−0.25	−0.25
SKINNY	15	105	−0.25	−0.25

7.5 Active-eigenspace curvature

The curvature diagnostic of Section 6 is evaluated in the 15-dimensional active eigenspace of the boundary covariance matrix. The result is constant for all coordinate planes in the tested cases.

7.6 Reproducibility checklist

The computational claims in this section are reproducible from the additional files. The scripts `gen_figures.py`, `compute_curvatures.py`, and `verify_curvature.py` regenerate the figures, active covariance diagnostics, and curvature checks. The rank/eigenvalue statement is proved independently in Theorem 6.1; the scripts provide finite checks for the reported 4-bit examples.

8 Discussion

8.1 What the framework establishes

The framework establishes a controlled information-geometric model for the local S-box distributions that appear in side-channel key recovery. Its first role is structural: exact key-induced distributions are singular boundary points, while ordinary Fisher–Rao geometry belongs to full-support interiors or to the adversary’s belief simplex. This distinction prevents assigning non-degenerate Fisher metrics, finite KL divergences, or ordinary geodesics to disjoint-support key endpoints.

The second role is algebraic. Walsh characters give explicit coordinates for the boundary orbit, and the key-translation rule determines how expectation coordinates move when the key hypothesis changes. This rule is elementary, but it is useful because it gives closed-form covariance and displacement expressions.

The third role is security-oriented. CPA is a Bayesian trajectory on the belief simplex. Under the Gaussian leakage model, pairwise Hamming-weight gaps determine expected log-Bayes gains against wrong keys. The scalar leakage variance is therefore a baseline descriptor, whereas pairwise gaps describe the discrimination structure relevant to key recovery.

This gives a concrete review criterion for the proposed quantities. A diagnostic is useful for side-channel security only if it changes the interpretation of posterior-odds growth, hard key differences, or the validity of a geometric approximation. The numerical section therefore does not stop at showing that PRESENT, GIFT, and SKINNY share Walsh-amplitude invariants; it also reports the pairwise gap distributions that determine which wrong keys are hardest to eliminate under the stated CPA model.

8.2 What is not claimed

The paper does not provide a new classification of 4-bit S-boxes. The classification and equivalence theory of small vectorial Boolean functions is an established subject. PRESENT, GIFT, and SKINNY are used as reproducible lightweight examples, not as evidence that the proposed spectral quantities separate S-boxes with the same Walsh amplitude distribution.

The paper also does not propose a new side-channel attack or countermeasure. The CPA model is the standard Gaussian Hamming-weight model. The contribution is the geometric separation of the boundary S-box law from the posterior belief law and the identification of which quantities are forced by bijectivity.

1427 8.3 Cryptographic interpretation

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1429 For isolated bijective 4-bit S-boxes under uniform input and Hamming-weight leak-
1430 age, $I_S(\text{HW}, \sigma) = 1/\sigma^2$ for every bijection. This equality is not a ranking criterion
1432 for S-boxes. Practical side-channel behavior depends on pairwise leakage gaps, imple-
1433 mentation leakage, noise, masking, alignment, higher-order effects, and the full cipher
1435 context.

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1437 In this sense, identical Walsh-amplitude histograms are a useful negative control
1438 rather than a failure of the method. They show that amplitude-only Walsh diagnostics
1439 cannot distinguish the three tested S-boxes, while the CPA gap distribution still
1440 exposes model-dependent hard and easy key differences. This is precisely the separation
1442 required before using information-geometric language in a cybersecurity setting.

1445

1446 The boundary covariance and local Walsh/Fisher separation proxy complement
1447 LAT, DDT, and empirical leakage evaluation. They are best understood as coordinate
1448 diagnostics for the singular key orbit and as inputs to posterior belief dynamics, not
1449 as replacements for standard cryptanalytic criteria.

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1454 8.4 Status of the curvature finding

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1456 The active-eigenspace value $K = -1/4$ is reported as a reproducible diagnostic under
1457 the projection, normalization, and sign conventions specified in Section 6. The rank and
1459 active eigenvalue statement is proved for every bijective S-box. A complete symbolic
1461 characterization of the curvature diagnostic, including the exact hypotheses under
1462 which it is constant, remains open.

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1466 8.5 Open problems

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1468 Several extensions are natural. First, the $\varepsilon \downarrow 0$ limit of regularized Fisher–Rao geometry
1469 could be studied using tools from singular information geometry. Second, the active-
1470 eigenspace curvature phenomenon could be attacked symbolically. Third, the leakage

model could be extended to masked, higher-order, non-Gaussian, or profiled settings. 1473
 Fourth, full-round ciphers would require propagation through linear layers and key 1474
 schedules, which is outside the isolated S-box model considered here. 1475
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9 Conclusion 1483

We presented a boundary information-geometric framework for the local S-box dis- 1484
 tributions that arise in side-channel key recovery. The main modelling point is the 1485
 separation between three spaces: the regular full-support Walsh exponential family, 1486
 the singular key-induced boundary orbit, and the adversary’s belief simplex. This 1487
 separation makes Fisher, KL, and projection statements mathematically well scoped. 1488
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For bijective S-boxes, Walsh coordinates give closed-form boundary covariance 1494
 matrices. The active covariance subspace has dimension $2^s - 1$ and all active eigenvalues 1495
 equal 2^s in the unnormalised Walsh basis. Thus this part of the geometry is a general 1496
 structural consequence of bijectivity. 1497
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For CPA, Bayesian updates occur on the belief simplex. Under Hamming-weight 1501
 Gaussian leakage, the scalar variance information is a baseline fixed by bijectivity, 1502
 while pairwise leakage gaps determine expected posterior-odds growth against wrong 1503
 keys. Computations on PRESENT, GIFT, and SKINNY illustrate this distinction and 1504
 reproduce the active curvature diagnostic $K = -1/4$ in the tested 4-bit cases. 1505
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The framework is not a new S-box classification and not a standalone attack. 1509
 Its contribution is a controlled geometric language for connecting singular S-box 1510
 distributions, boundary covariance, and side-channel Bayesian inference. The security 1511
 value of the framework is to make explicit which geometric quantities are structural 1512
 invariants and which quantities affect local key-hypothesis discrimination under a 1513
 specified leakage model. 1514
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1519 **Declarations**

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1521 **Availability of data and materials**

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1524 The datasets supporting the conclusions of this article are included within the arti-
1525 cle and its additional files. Additional file 1, `supplementary_reproducibility.pdf`,
1526 records the computational protocol, software environment, arithmetic conventions, and
1527 reproducibility coverage. Additional file 2, `supplementary_material.zip`, contains
1528 the scripts and dependency list used to regenerate the numerical tables and figures.

1532 Software information: project name, Boundary Information Geometry for S-box
1533 Side-Channel Leakage; project home page, [https://github.com/EkodeckStephane/](https://github.com/EkodeckStephane/IG-based-Sbox-SCL)
1534 `IG-based-Sbox-SCL`; archived version, GitHub tag v1.0.4, [https://github.com/](https://github.com/EkodeckStephane/IG-based-Sbox-SCL/archive/refs/tags/v1.0.4.zip)
1535 `EkodeckStephane/IG-based-Sbox-SCL/archive/refs/tags/v1.0.4.zip`; operating system,
1536 platform independent; programming language, Python 3; other requirements, `numpy`,
1537 `scipy`, and `matplotlib`; license, not specified; restrictions to use by non-academics,
1538 none beyond the repository license status.

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1545 **Competing interests**

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1547 The authors declare that they have no competing interests.

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1554

1555 **Authors' contributions**

1556

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1558 S.A.E., P.R.M.A., and R.N.; software and reproducible computations: S.G.R.E.;
1559 validation: S.A.E., P.R.M.A., and R.N.; supervision and project administration:
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